

Newton-Raphson

$$f(x) = x^3 - 10x^2 + 5, \quad f'(x) = 3x^2 - 20x$$

$$f(x_1) \approx f(x_0) + f'(x_0)(x_1 - x_0) = 0$$

$$x_0 = 1, \quad f(1) = -4, \quad f'(1) = -17$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{-4}{-17} = \frac{13}{17} \approx 0.7647058824$$

In [1]: `import pandas as pd`

```
f = lambda x: x**3 - 10*x**2 + 5
f_prime = lambda x: 3*x**2 - 20*x

x0 = 1
x1 = x0 - f(x0)/f_prime(x0)
print('First approximation:', x1)
```

First approximation: 0.7647058823529411

In [2]: `def newton_raphson(f, f_prime, x0, tol=1e-6):`

```
    x1 = x0 - f(x0)/f_prime(x0)
    n = 1

    while abs(x1 - x0) > tol:
        x0 = x1
        x1 = x0 - f(x0)/f_prime(x0)
        n += 1

    return x1, n

print(newton_raphson(f, f_prime, x0))
```

(0.7346035077893185, 4)

In [3]: `reference = 0.7346035077893032`

```
rows = []
tolerances = []

for i in range(1, 6):
    tol = 10**(-i)
    root, steps = newton_raphson(f, f_prime, x0, tol)
    error = abs(reference - root)/reference*100
    tolerances.append(tol)
    rows.append([root, steps, error])

columns = ['Root', 'Steps', 'Error %']
pd.DataFrame(rows, index=tolerances, columns=columns)
```

Out[3]:

	Root	Steps	Error %
0.10000	0.735121	2	7.047776e-02
0.01000	0.734604	3	2.174372e-05
0.00100	0.734604	3	2.174372e-05
0.00010	0.734604	4	2.085625e-12
0.00001	0.734604	4	2.085625e-12